

Track A - Privacy, Trust, and Responsible Agentic Systems

#1 Invisible Compliance: Privacy-by-Design in the Developer Workflow

As software teams move fast, privacy requirements are often reviewed too late, after features are already built, data flows are already established, and user-facing promises may already be inconsistent with actual system behaviour. This challenge invites participants to build **AI-agentic systems** that embed privacy thinking directly into the developer workflow, making compliance a continuous, actionable, and developer-friendly process rather than a last-minute legal review.

Problem Statement

In many software projects, privacy is still treated as a late-stage review or a legal checkpoint rather than something built into development from the start. Developers may add new data fields, third-party APIs, analytics tools, or logging mechanisms without fully understanding the privacy impact of those changes. This creates a gap between what the code does and what privacy policies, consent flows, or compliance documents promise to users.

The result is that privacy often becomes reactive, expensive, and difficult to maintain. Developers need tools that can bring privacy awareness directly into their everyday workflow.

Challenge Scope

Build an **agentic system** that helps developers integrate privacy and compliance into the natural software development process.

Your solution should be able to interpret technical changes such as code diffs, schema updates, new APIs, or third-party integrations, and provide immediate privacy-related feedback. It may also help update documentation, flag potential compliance issues, or recommend privacy-preserving design choices.

The goal is to make **Compliance-by-Design** the default state, so developers can make privacy-aware decisions without needing to be legal experts.

Potential Solution Areas

Projects may include, but are not limited to:

- **Code diff privacy analysers** that detect privacy-relevant changes in pull requests
- **Privacy-aware IDE assistants** that give real-time feedback during development
- **Third-party API / SDK review agents** that identify new data-sharing risks
- **Automated privacy documentation tools** that update notices, data inventories, or internal records
- **CI/CD privacy guardrails** that check compliance risks before deployment
- **Data flow mapping tools** that show how personal data moves through a system
- **Developer-facing explainers** that translate privacy obligations into engineering actions

#2 Respectful Automation: Privacy Personas for Autonomous AI

Problem Statement

As AI systems become more autonomous, they are increasingly able to complete tasks on behalf of users, such as booking travel, managing schedules, or interacting with online services. However, these systems often optimise for efficiency and task completion without fully respecting user privacy boundaries. An agent may complete a task while unnecessarily exposing sensitive information, such as personal preferences, location history, financial details, or behavioural patterns.

The core issue is that autonomy without privacy awareness can lead to actions that are effective, but not respectful. Users need AI systems that do not just act for them, but act **within their comfort level and personal boundaries**.

Challenge Scope

Build a framework, agentic governor, or intelligent control layer that helps autonomous systems operate according to **user-defined privacy personas**.

A privacy persona represents the user's preferred level of data sharing, disclosure, and automation. Your solution should help agents make decisions such as:

- What information can be accessed for a task?
- What data should remain hidden unless necessary?
- Which third parties are acceptable to interact with?
- When should the agent ask for confirmation before proceeding?
- How should the agent adapt its behaviour for users with different privacy comfort levels?

The goal is to create AI systems that balance **usefulness, autonomy, and respect for user privacy boundaries**.

Potential Solution Areas

Projects may include, but are not limited to:

- **Privacy persona engines** that define user comfort levels, such as conservative, balanced, or convenience-first
- **Agentic governors** that restrict what an autonomous agent can access, share, or do
- **Context-aware permission systems** that adjust decisions based on task sensitivity
- **Confirmation layers** that determine when an agent should ask before taking action
- **Privacy-aware travel, shopping, or scheduling agents** that minimise unnecessary disclosure
- **Simulation tools** that test whether an agent's behaviour aligns with a user's privacy persona
- **Explainability tools** that show why the agent decided to share, hide, or request data

Track B - AI for Education and Campus Experience

#1 Responsible Learning: Transparent and Ethical AI in Education

Problem Statement

Generative AI is increasingly used by students for coding, writing, debugging, and problem-solving. However, many educational settings lack practical ways to document how AI was used, distinguish student effort from AI assistance, and assess whether students truly understand their submitted work. This creates ethical concerns around academic integrity, fairness, transparency, and learning outcomes.

In this challenge, teams are invited to design tools or systems that make AI-assisted learning more transparent, accountable, and educationally meaningful.

Challenge Scope

Design an AI-powered educational solution that helps students and instructors manage the ethical use of AI in learning. The solution should support transparency, attribution, responsible assistance, and fair assessment, especially in scenarios where students use AI tools for coding, writing, problem-solving, or project development.

Potential Solution Areas

Projects may include, but are not limited to:

- **AI Usage Disclosure & Documentation** - Build tools that help students clearly disclose how they used AI during an assignment.
- **Code Attribution & Contribution Labelling** - Design systems that identify, annotate, or document which parts of a coding assignment were written by the student, suggested by AI, or modified from AI output.

- **Assignment Policy Interpretation** - Create tools that help students understand what kinds of AI use are allowed for a specific course or assignment.
- **Learning Process Tracking** - Build systems that capture the student's learning process rather than only evaluating the final submission.

#2 AI for Campus Life: Solving the Missing Piece

Problem Statement

Every student's time on campus includes small but persistent frustrations: difficulty finding the right resources, missing important deadlines, struggling to navigate academic systems, feeling disconnected from campus opportunities, or facing everyday inefficiencies that no single office or app fully solves. These issues are often not dramatic enough to be treated as major institutional problems, but they meaningfully affect student experience, belonging, productivity, and well-being.

The challenge is to reflect on what felt missing in campus life and build an AI solution that addresses that gap in a practical, student-centred way. The goal is not necessarily to create a highly agentic system, but to use AI meaningfully to improve the everyday campus experience.

Challenge Scope

Build an **AI-powered solution** that solves a real friction point, unmet need, or overlooked gap in campus life.

Your project can focus on any part of the student experience, such as:

- Academics and course planning
- Navigating university services
- student well-being and support
- campus events and opportunities
- community building and belonging
- communication overload and information access
- productivity, scheduling, or decision support

The key is to identify something that felt missing, confusing, inefficient, or unsupported during campus life, and design a solution that makes that experience better.

Potential Solution Areas

Projects may include, but are not limited to:

- **Campus resource assistants** who help students find the right office, support service, or policy
- **Opportunity discovery platforms** for clubs, research, internships, events, or scholarships
- **Student well-being tools** that reduce stress, confusion, or isolation
- **Smart campus navigation or information systems** that simplify day-to-day campus life
- **AI-powered communication organisers** that summarise emails, announcements, and deadlines
- **Community and connection platforms** that help students find peers, mentors, or relevant groups

Technical Rules

1. Disclosure of Models, Datasets, and External Services

Teams must disclose all major technical components used in their project, including foundation models, datasets, APIs, third-party services, and external tools. If a dataset is used, teams should clearly state its source, whether it is public, private, synthetic, or self-collected, and how it was processed.

2. Data Provenance and Usage Declaration

Any team using external or pre-existing datasets must explain where the data came from, what permissions or licenses apply, and whether any personal, sensitive, or campus-related data is included. Teams should not imply that a dataset is original if it is adapted from another source.

3. Privacy and Sensitive Data Restriction

Teams must not use real sensitive personal data unless they have explicit authorisation and can clearly explain how privacy is protected. Whenever possible, teams should use anonymised, synthetic, or demo-only data.

4. AI-Generated Content Disclosure

If teams use generative AI tools for coding, design, content generation, prototyping, or presentation materials, they should disclose that use at a high level. Disclosure will not negatively affect judging, but undisclosed heavy reliance may be penalised if it misrepresents the team's own work.

5. Reproducibility / Demo Transparency

Teams should be prepared to explain what parts of the system are fully implemented, what parts are mocked or simulated, and what assumptions were made in the demo. Simulations are allowed, but they must be clearly labelled.

6. No Misleading Claims

Teams should not make unsupported claims such as “fully compliant,” “guaranteed privacy-safe,” “detects all misuse,” or “bias-free” unless they can substantiate them. Judges may penalise overclaiming.